

09 – Databases Recap

What we studied and learned in this course

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DATABASES RECAP

- COLLECTION OF DATA (WITH DATATYPES).
- KEEP / STORE INFORMATION.
- USE INFORMATION.

DATABASES TYPES

SQL

No SQL

- Structured → Slow
Data Manipulation Language

- (Not) structured → fast
- XML or JSON or ...

Data Definition

DDL

CREATE DATABASE
CREATE TABLES
ALTER ...
CREATE INDEX
...

DML

(CRUD)

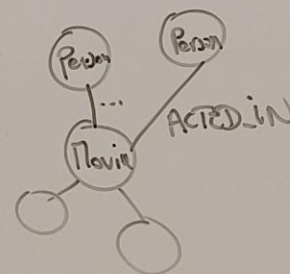
CREATE READ UPDATE DELETE

POST GET PUT DELETE

REST API

SELECT * FROM ...;
INSERT INTO ... (...) VALUES (...); (...), (...);
UPDATE
DELETE

NEO4J



MATCH...

Mongodb

db.collection ...
[{id: 1, "fname": "Jerran"},
 {id: 2, "fname": "Mira"}
]

db.collection.find({...})
db.collection.aggregate([...])

JSON JavaScript Object Notation

7 characters:

[...] : Array

{ ... } : Object

"..." : String

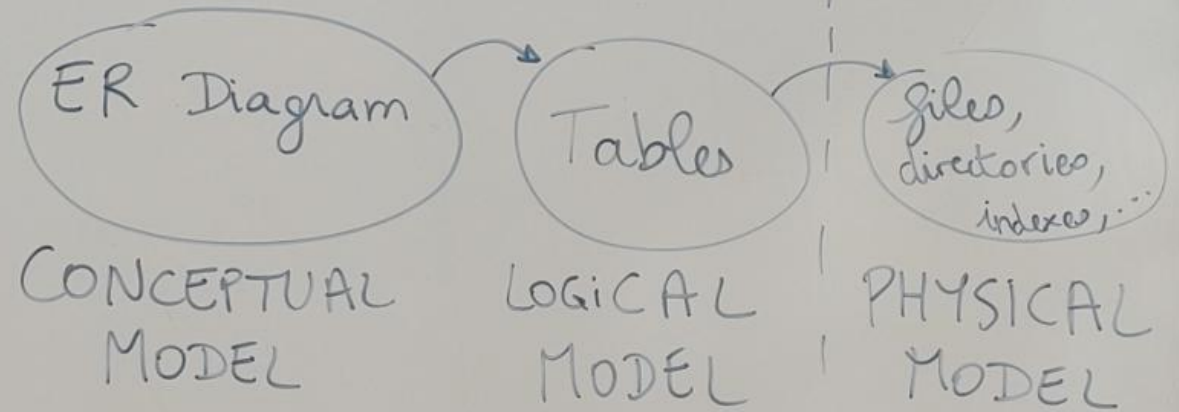
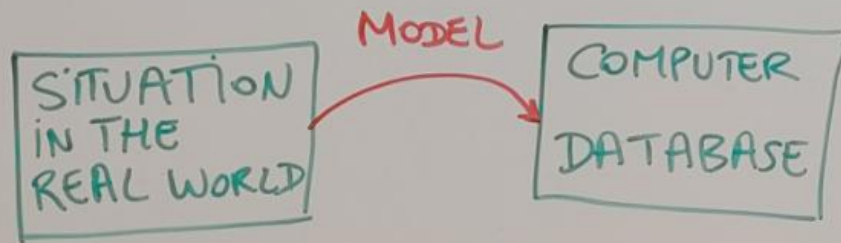
"key" : "value" : separator key/value

, : separator between 2 objects
2 key/value pairs

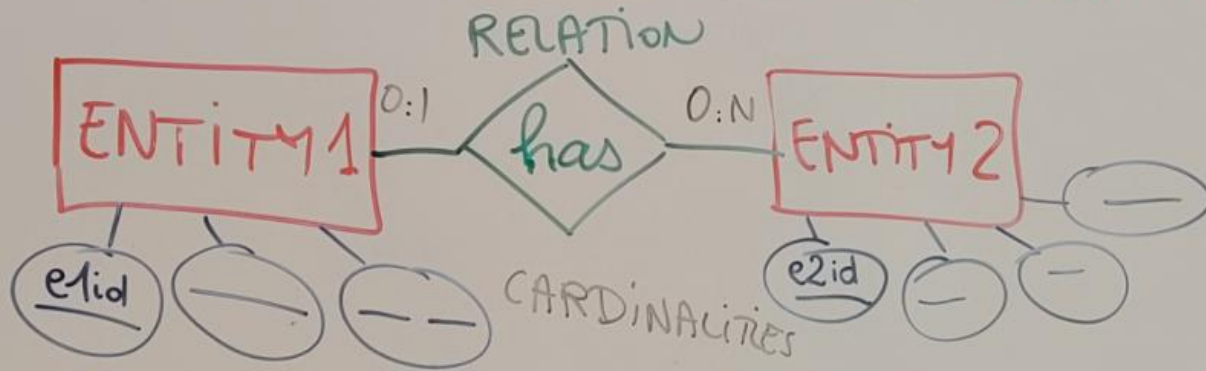
let countries =

```
[ { "id": 1, "name": "SWEDEN", "population": 11500000,
  "capital-city": "Stockholm" }, { "id": 2, "name": "IRELAND",
  "capital-city": "Dublin", "language":
  [ { "name-l": "Irish" }, { "name-l": "English" } ] },
  { "id": 35, "name": "FRANCE", "capital-city":
    "Paris", "population": 68300000 } ]
```

MODELIZATION



ENTITY-RELATIONSHIP DIAGRAM

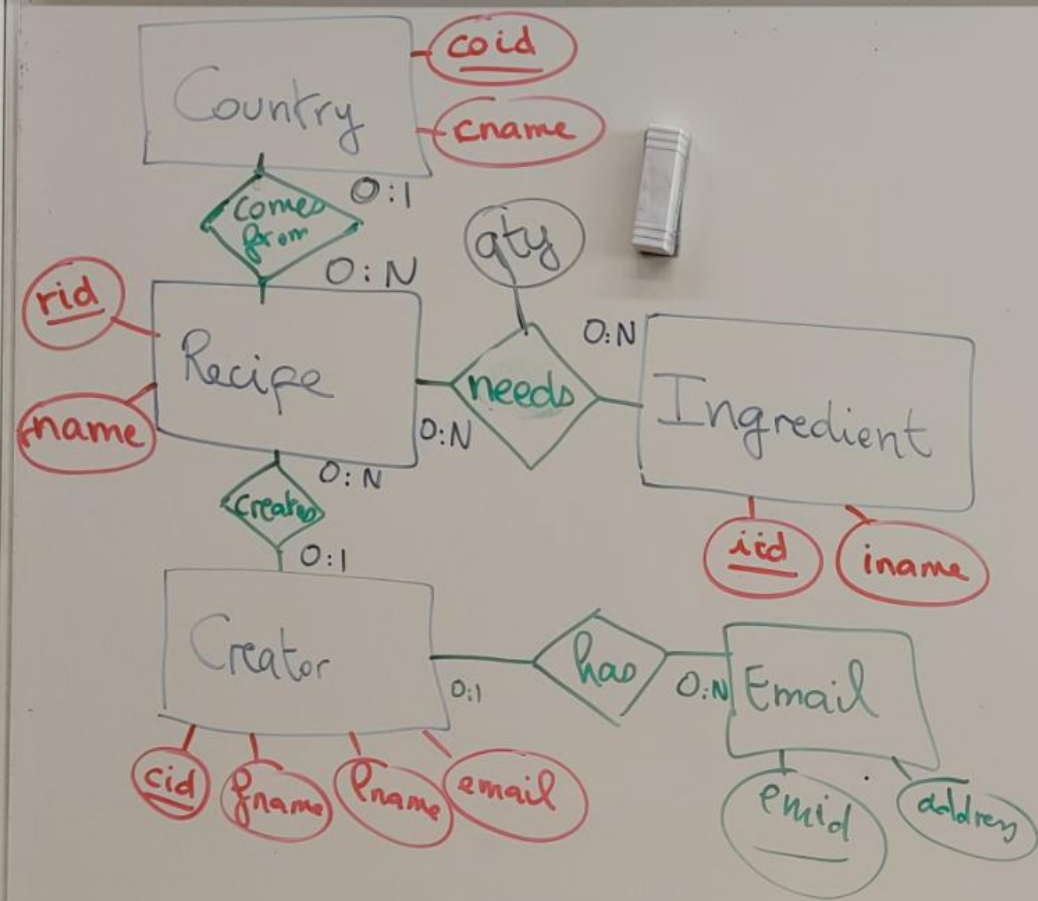


RULES

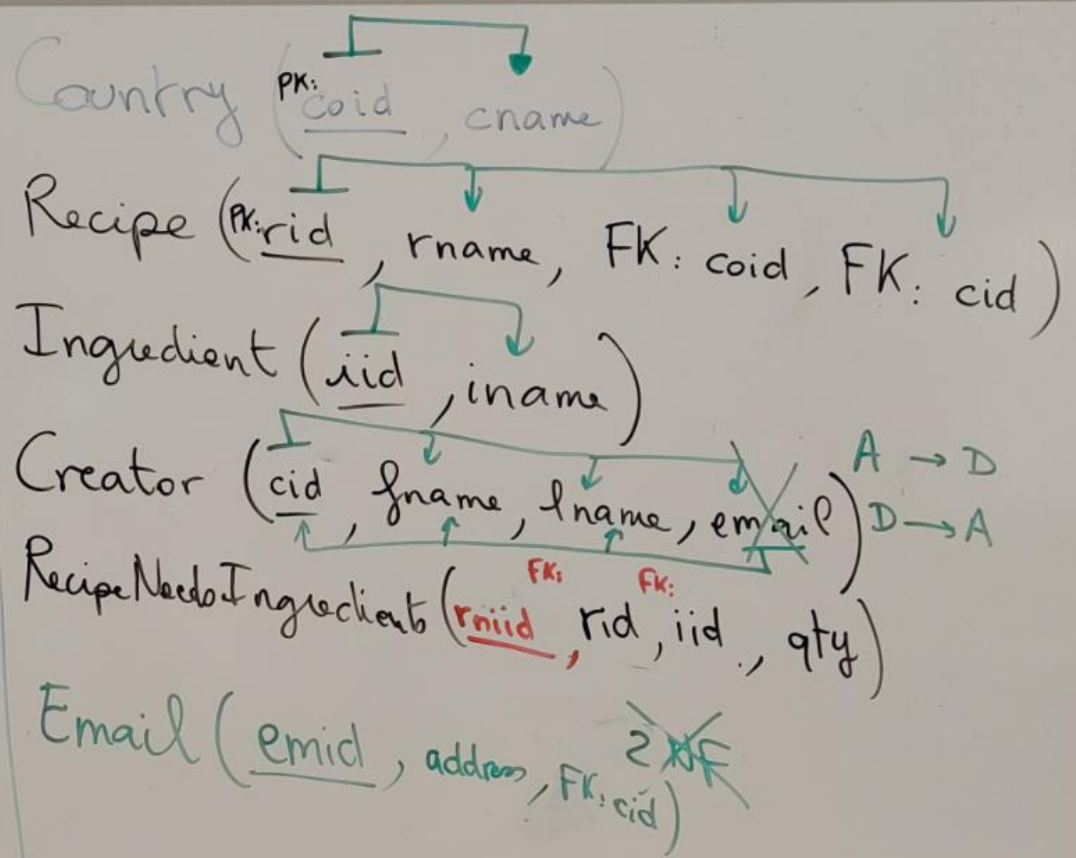
- 1) EVERY ENTITY BECOMES A TABLE
- 2) CARDINALITIES:
1:N
 - a) The PK on the 1 side becomes a FK in the table on the N side.
 - b) Many-to-many relations become a 'full' table

ATTRIBUTES

LOGS
ENTITY 3

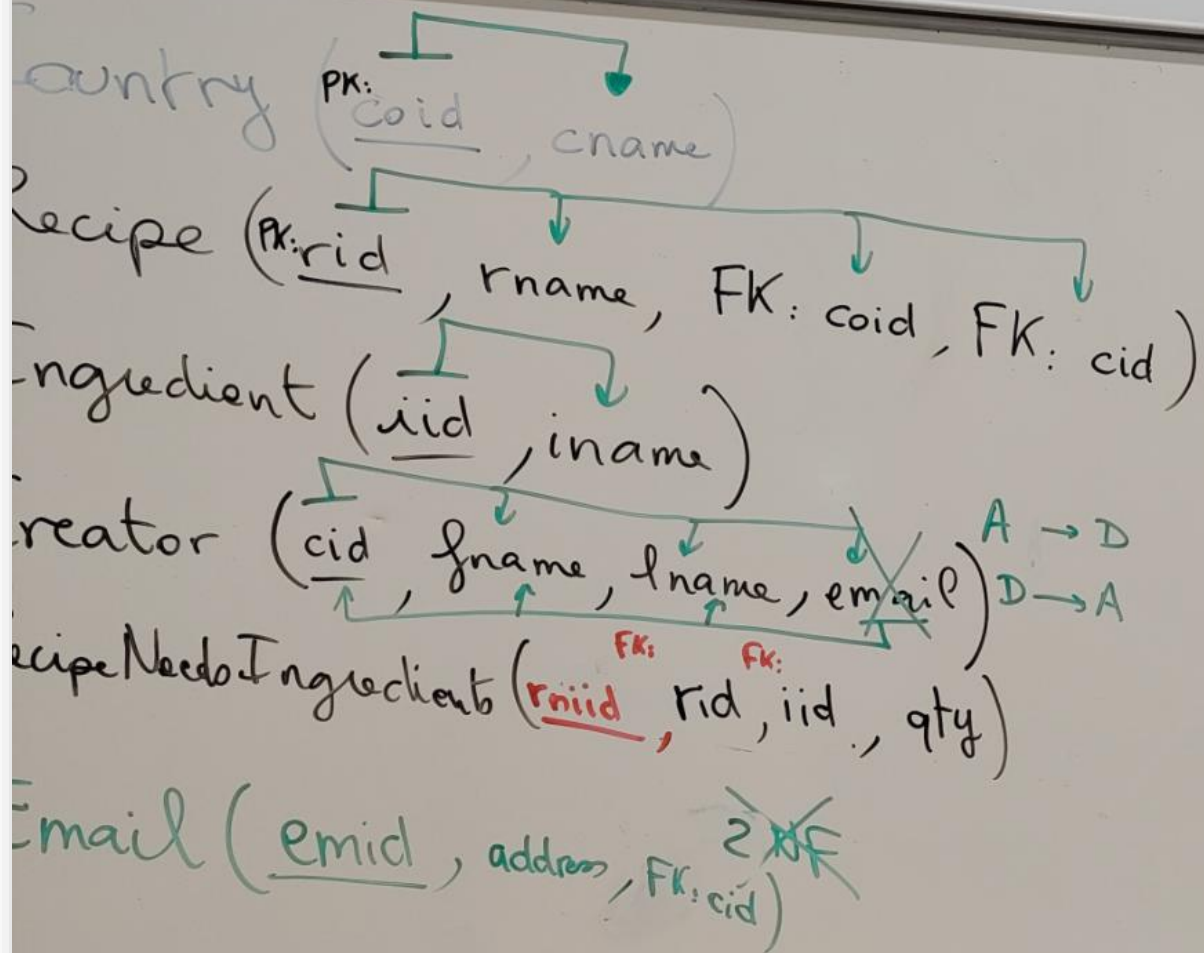


CONCEPTUAL MODEL (ERD)



LOGICAL MODEL (JDEL) (TABLES)

No
1N
2N
3N



NORMALIZATION

1NF: ONLY ONE VALUE PER CELL
NO REPEATING GROUPS

2NF: 1NF

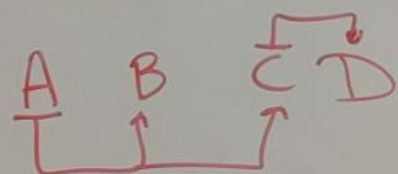
ONE COLUMN AS A PRIMARY KEY

3NF: 2NF

NO TRANSITIVE FUNCTIONAL DEPENDENCIES

LOGICAL MODEL (TABLES)

CLOSURE



$$\{A\}^+ = \{A, B, C, D\}$$

$$\{B\}^+ = \{B\}$$

$$\{C\}^+ = \{C, D\}$$

$$\{D\}^+ = \{D\}$$

JOINS

Creator

(1, "Jerome", "Lande")
(2, "Mira", "Pop")

Recipe

(1, "pizza 1", 7, 1)
(17, "meatballs", 2, 2)

SELECT * FROM Creator

JOIN Recipe

ON Creator.cid = Recipe.cid;

cartesian product

Creator $\times$ Recipe = [

{1, "Jerome", "Lande", 1, "pizza 1", 7, 1}

~~{1, "Jerome", "Lande", 17, "meatballs", 2, 2}~~

~~{2, "Mira", "Pop", 1, "pizza 1", 7, 1}~~

~~{2, "Mira", "Pop", 17, "meatballs", 2, 2}~~

]

ANY QUESTIONS?

BE POSITIVE, B+!



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